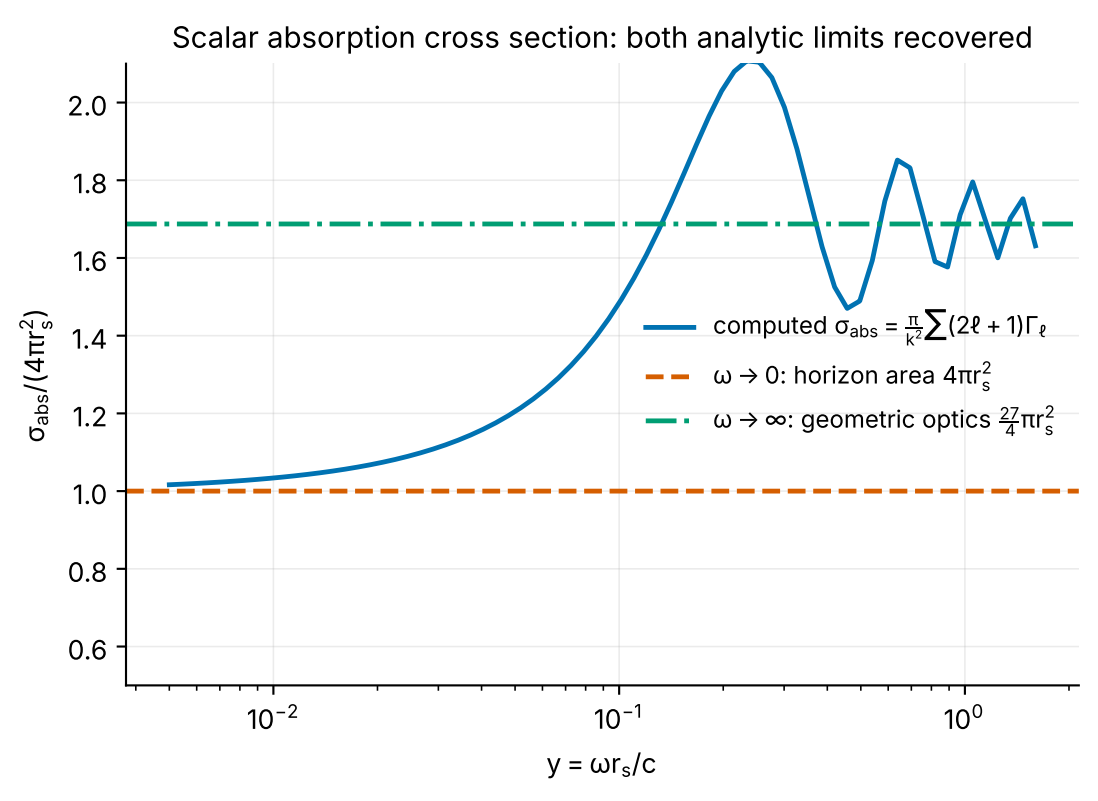
Scalar absorption cross section: both analytic limits recovered
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
$\sigma_{abs}/(4\pi r_s^2)$
computed $\sigma_{abs} = \frac{\pi}{k^2}\sum(2\ell + 1)\Gamma_\ell$
$\omega \to 0$: horizon area $4\pi r_s^2$
$\omega \to \infty$: geometric optics $\frac{27}{4}\pi r_s^2$
10^{-2}
10^{-1}
10^{0}
$y = \omega r_s/c$